

AI OPs Assignment 1 Question 1

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1. a. This is a case of **2. Entanglement (CACE)**.
This is clearly a case where changing one thing randomly changes something else or changes a lot of other things. Parts that are supposed to be separate, are not actually separate due to underlying lack of maintenance.
 - b. This is a case of **4. Undeclared Consumer**.
This is risky because when a team silently consumes another team's work, they might end up accidentally relying on it. Since the original team (ML team) doesn't know about it, they might end up pushing changes that break the marketing team's use case of the table.
 - c. This is a case of **6. Configuration & Glue-Code Debt**.
This is a case of a pipeline jungle where no one can clearly figure out where things are. It might have been easy for the creators to set up, and might represent the chronology of how the feature developed naturally, but is terribly difficult to maintain.
2.
 - The third case can be improved using the Git + DVC + MLflow pipeline that we studied in class.
 - Instead of having 14 undocumented shell scripts, we should have one clear Jupyter notebook that flows through all the processes of all the 14 scripts. At every stage, we should have an MLflow wrapper so that everything is automatically logged and tracked. MLflow will automatically take care of tracking all the hyperparameters.
 - If there are multiple versions of the data, we should use DVC as needed.